

REMORA: Real-time Expressive Motion to Output Routing Audio

A Martial Arts Staff Instrument Driven by Motion

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ABSTRACT

REMORA turns a bō, the long staff used in martial arts, into a musical instrument: spins, tilts, and strikes become sound in real time. A small wireless sensor rides on the staff and reads its motion continuously, translating qualities borrowed from Laban Movement Analysis (how forcefully, suddenly, and freely the staff moves) into pitch, texture, and timbre across three very different synthesis voices, from a warm additive choir to a breath-driven vocal tract. Fifteen presets span a range from a beginner's tilt-controlled pitch to compound mappings that reward a martial artist's own staff-handling vocabulary.

1. INTRODUCTION

REMORA grew out of the first author's parallel practice in martial arts and music. The bō itself is a form largely absent from new interfaces for musical expression: a full-body, two-handed, swung staff with its own weight, reach, and rotational inertia, quite unlike the small handheld or body-worn sensors most inertial digital instruments use. Where a martial artist reads speed, force, and reach off the staff by feel, REMORA reads the same qualities electronically and turns them into sound: a spin becomes a filter sweep, a strike becomes a percussive transient, a slow controlled turn sounds nothing like a fast, loose one. The instrument's vocabulary is, in that sense, borrowed directly from the performer's own body of practice rather than invented for the occasion.

Rod-, staff-, and baton-shaped controllers already form a small, recurring family in this space: the T-Stick's touch-sensitive shaft [1], the Digital Baton's conducting gesture [2], and Nam's swung Musical Poi [3] among them. What REMORA adds to that family is less a new sensor than a new relationship: a martial artist's years of staff-handling skill, carried straight onto a stage as a musical vocabulary.



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(a) Parametric design (c) Hardware wiring (b) Printed and mounted

Figure 1: The staff-mounted sensor unit, held on with zip ties so it can be repositioned or removed without marking the staff: (a) the parametric OpenSCAD design; (b) the printed enclosure zip-tied to the bō.

2. BUILDING THE INSTRUMENT

Inside the enclosure shown in Figure 1, a microcontroller reads a nine-axis inertial sensor and streams orientation, acceleration, and gyroscope data wirelessly to a laptop at 100 Hz. Nothing tethers the performer to a fixed point on stage; a cable is needed only to charge the unit between sets.

Because the sensor can be glued to the staff at any angle (exact placement is never perfectly reproducible by hand, and often changes from one build of the enclosure to the next), the plugin calibrates itself the first time the staff is held through three reference poses, working out which physical axis is “forward” and correcting for any twist around it. From then on, the same instrument logic runs correctly no matter where on the staff the sensor happens to sit, so a performer can move it without re-learning the instrument underneath.

3. HOW MOTION BECOMES SOUND

Every packet from the sensor is read against a moving reference: not just where the staff is, but how it got there. Four continuous qualities borrowed from Laban Movement Analysis are extracted from the motion stream every audio block. How forcefully the staff moves becomes Weight, read from a leaky integral of its acceleration once gravity is subtracted out, and typically driving loudness or waveshaping. How suddenly a spin speeds up or stops becomes Time, driving attack speed and how quickly a chord change lands. Whether the staff keeps spinning around the same axis or wanders becomes Space, a timbral focus that tightens or loosens a filter or a chorus spread. How bound or free the motion is becomes Flow, crossfading between a controlled and a freer, noisier version of the same sound. Tilt and roll bend pitch and blend voices continu-

ously; a spin’s lap count sweeps a filter cutoff along a slow curve; a stab or strike, a sharp reversal in the staff-tip’s acceleration, triggers a transient, a chord change, or a percussive burst.

What these qualities actually sound like depends on which of three voices they are driving. A default additive engine sums four voices of six phase-locked harmonics each, vibrato- and chorus-detuned into a warm, faintly unstable choir that a mapping can drive into a rasping saturation or filter down to a single dark hum; a short comb resonator gives every voice a string-like colouration. A granular engine synthesises its own two-second bed of shifting harmonics and filtered noise, then scatters up to 64 overlapping grains across it (forward, backward, panned, re-pitched) for a windy, textural voice with no fixed pitch centre at all, currently reserved for the *Wind Noise* preset. A third engine drives a digital-waveguide vocal tract excited by a glottal-pulse model and breathed into with filtered noise, so the instrument can also simply speak, or half-speak, rather than only sing.

Internally, which motion quality drives which sound parameter is authored as a small patch of connected building blocks inside the plugin (Figure 2), on a visual canvas in the lineage of Max [4] and Pure Data [5]. That patching layer is a tool for building and tuning the instrument.

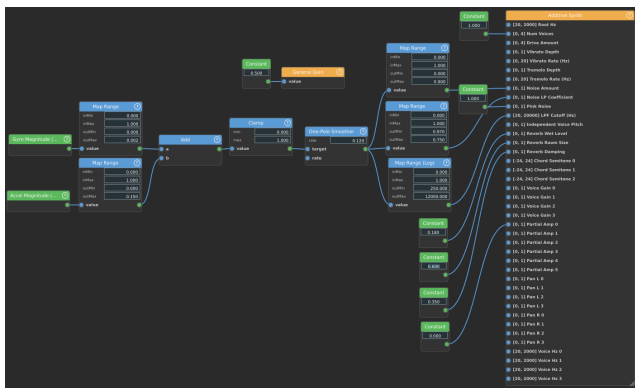


Figure 2: One preset, seen inside the plugin: motion sources (green) run through shaping and smoothing math (blue) into the parameters of an additive-synthesis voice (orange).

4. A VOCABULARY OF PRESETS

Fifteen named presets make up the instrument’s current repertoire, graded by the physical vocabulary they ask of a performer rather than uniform in difficulty. *Simple* asks only for basic tilt and roll, an easy way in that needs no martial background at all. *Martial* and *Azimuth* reward the staff-handling skill a martial artist already carries onto a stage: they derive a facing direction and a spin quality straight from the calibrated motion stream, without ever touching a magnetometer (which a stage rig’s own electronics would only confuse), so that where the staff points and how it turns becomes part of the music. *Wind Noise* trades pitch for a breathy, granular texture sustained by nothing but motion; *Vocal Tract* pushes the same sensor stream into something closer to a voice than a synthesiser. Each preset is less a separate program than a different character the same instrument can take on, chosen for a set the way a performer chooses which piece to play next, rather than a fixed, single-piece score.

5. WHERE THIS STANDS

Everything above is grounded in the system as it runs today, exercised continuously through the first author’s own development and rehearsal. What it is not yet grounded in is anyone else’s experience of the instrument: no performer besides its designer has sat down with the staff, and the claims above should be read as implementation-level evidence, not validated usability. A semi-structured session with performers who have some staff, baton, or movement background (but no prior exposure to REMORA) is planned next: a calibration walkthrough, free play across a small span of presets, and a closing interview, aimed squarely at the question this abstract cannot yet answer: whether the instrument is playable, and its motion vocabulary legible, to someone other than the person who built it.

The instrument also carries limitations worth naming plainly: Wi-Fi pairing is still a manual, reflash-to-change affair rather than a runtime setting; yaw drift is bounded to sessions of tens of minutes by assumption rather than measurement; and the single inertial sensor gives REMORA no sense of a second performer, so it remains a solo instrument for now.

6. ACKNOWLEDGMENT

This work was carried out during the first author’s internship at CREATE, Aalborg University, Denmark. Firmware, plugin source, enclosure files, and all fifteen presets are available at <https://github.com/TimothéeDvt/REMORA>.

7. REFERENCES

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